

VECM Summary from RW4 Primary Chain

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

RW4 domestic vertical chain is FarmGateUA -> ProducerUA -> ProZorro -> Retail, estimated with both forward and reverse-flow pairs. Farm-gate enters from two alternative reconstruction workbooks and both linear/pchip interpolation variants are carried end-to-end.

family=VECM

rows=156

source=/Users/getapple/Documents/KSE/Master Thesis/Main materials/Model/Charniuk_Dairy_Research/outputs/primary_chain_summary/primary_chain_consolidated.xlsx

Table: VECM_Summary

panel_level	panel_name	segment_name	product	standardized_type	retailer_panel	brand	price_variant	reconstruction_variant	farm_gate_source	frequency	chain_direction	stage_from	stage_to	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	unreliable_reason	shock_dummy_included	promo_controls_included	ljungbox_p	arch_p	jb_p	notes	robust_linear_vs_pchip	interpolation_sensitive	robust_across_reconstruction	core_finding_flag
product	product::Вершки::Silpo::o...	Вершки	Вершки	cream	Silpo	nan	observed	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	41	nan	nan	2.214812140355627	nan	nan	nan	2.0	ok	0	nan	0	0.9397102432548599	0.785228977301169	0.02977676690767751	k_ar_diff=5	0	1	0	0	
product	product::Молоко питне::Sil...	Молоко питне	Молоко питне	milk	Silpo	nan	observed	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	41	nan	nan	2.542086532312744	nan	nan	nan	2.0	ok	0	nan	0	0.9910630862290195	0.9998696776320517	3.549057433899505e-160	k_ar_diff=5	0	1	0	0	
product	product::Сир твердый::Sil...	Сир твердый	Сир твердый	hard_cheese	Silpo	nan	observed	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	38	nan	nan	-3.281200511143566	nan	nan	nan	3.0	ok	0	nan	0	0.9908513112556221	0.5717024280852399	0.1445401016754244	k_ar_diff=5	0	1	0	0	
product	product::Сметана::Silpo::b...	Сметана	Сметана	sour_cream	Silpo	nan	observed	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	38	nan	nan	-24.70966978176242	nan	nan	nan	2.0	ok	0	nan	0	0.1646681594712951	0.9828120691897677	0.9191199356810634	k_ar_diff=5	0	1	0	0	
product	product::Вершки::Silpo::b...	Вершки	Вершки	cream	Silpo	nan	baseline	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	41	nan	nan	2.214812140355627	nan	nan	nan	2.0	ok	0	nan	0	0.9397102432548599	0.785228977301169	0.02977676690767751	k_ar_diff=5	0	1	0	0	
product	product::Молоко питне::Sil...	Молоко питне	Молоко питне	milk	Silpo	nan	baseline	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	41	nan	nan	2.542086532312744	nan	nan	nan	2.0	ok	0	nan	0	0.9910630862290195	0.9998696776320517	3.549057433899505e-160	k_ar_diff=5	0	1	0	0	
product	product::Сир твердый::Sil...	Сир твердый	Сир твердый	hard_cheese	Silpo	nan	baseline	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	38	nan	nan	-3.281200511143566	nan	nan	nan	3.0	ok	0	nan	0	0.9908513112556221	0.5717024280852399	0.1445401016754244	k_ar_diff=5	0	1	0	0	
product	product::Сметана::Silpo::b...	Сметана	Сметана	sour_cream	Silpo	nan	baseline	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	38	nan	nan	-24.70966978176242	nan	nan	nan	2.0	ok	0	nan	0	0.1646681594712951	0.9828120691897677	0.9191199356810634	k_ar_diff=5	0	1	0	0	
average	average::Silpo::observed::...	all_products_average	all_products_average	all_products_average	Silpo	nan	observed	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	47	nan	nan	-26.52676975899912	nan	nan	nan	1.0	ok	0	nan	0	0.2270834451270314	0.6453889741668349	2.80087444612259e-10	k_ar_diff=5	0	1	0	0	
average	average::Silpo::observed::...	all_products_average	all_products_average	all_products_average	Silpo	nan	observed	linear	initial	daily	forward	ProducerUA	Retail	producer_prozorrr_retail_...	VECM	retail	producer+prozorro	47	nan	nan	0.00418690695050269	nan	nan	nan	1.0	ok	0	nan	0	0.3968749311479061	0.9021252136581074	0.536356806653772	k_ar_diff=1	0	1	0	0	
average	average::Silpo::observed::...	all_products_average	all_products_average	all_products_average	Silpo	nan	observed	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	retail	farm_gate+producer+prozorro	47	nan	nan	-0.02072783869708889	nan	nan	nan	2.0	ok	0	nan	0	0.7566017709831934	0.4187563657102568	0.2466730860498608	k_ar_diff=5	0	1	0	0	
average	average::Silpo::baseline::...	all_products_average	all_products_average	all_products_average	Silpo	nan	baseline	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	47	nan	nan	-26.52676975899912	nan	nan	nan	1.0	ok	0	nan	0	0.2270834451270314	0.6453889741668349	2.80087444612259e-10	k_ar_diff=5	0	1	0	0	
average	average::Silpo::baseline::...	all_products_average	all_products_average	all_products_average	Silpo	nan	baseline	linear	initial	daily	forward	ProducerUA	Retail	producer_prozorrr_retail_...	VECM	retail	producer+prozorro	47	nan	nan	0.0006769520452472977	nan	nan	nan	1.0	ok	0	nan	0	0.5318867755109824	0.3009767974834293	0.2634721710363644	k_ar_diff=1	0	1	0	0	
average	average::Silpo::baseline::...	all_products_average	all_products_average	all_products_average	Silpo	nan	baseline	linear	initial	daily	forward	FarmGateUA	Retail	farmgate_producer_prozorrr...	VECM	retail	farm_gate+producer+prozorro	47	nan	nan	-2.751633856116992	nan	nan	nan	3.0	ok	0	nan	0	0.6439939894565483	0.6334170863398159	0.7990449055600132	k_ar_diff=5	0	1	0	0	
comparison	comparison::silpo_vs_novu...	silpo_vs_novus_observed	silpo_vs_novus_observed	comparison_panel	silpo_vs_novus_observed	nan	observed_comparison	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	49	nan	nan	144.9687138737506	nan	nan	nan	2.0	ok	0	nan	0	0.6775706216442879	0.4816846028746816	0.5440430058842177	k_ar_diff=5	0	1	0	0	
comparison	comparison::silpo_vs_novu...	silpo_vs_novus_observed	silpo_vs_novus_observed	comparison_panel	silpo_vs_novus_observed	nan	observed_comparison	linear	initial	daily	forward	ProducerUA	Retail	producer_prozorrr_retail_...	VECM	retail	producer+prozorro	49	nan	nan	0.4755853663965547	nan	nan	nan	3.0	ok	0	nan	0	0.516367697550039	0.9320062462573419	0.6527988475863971	k_ar_diff=1	0	1	0	0	
comparison	comparison::silpo_vs_novu...	silpo_vs_novus_observed	silpo_vs_novus_observed	comparison_panel	silpo_vs_novus_observed	nan	observed_comparison	linear	initial	daily	forward	FarmGateUA	Retail	farmgate_producer_prozorrr...	VECM	retail	farm_gate+producer+prozorro	49	nan	nan	-20.8364357794366	nan	nan	nan	3.0	ok	0	nan	0	0.5751923476300824	0.243806286561618	0.7060092731291543	k_ar_diff=5	0	1	0	0	
comparison	comparison::silpo_baselin...	silpo_baseline_vs_novus_o...	silpo_baseline_vs_novus_o...	comparison_panel	silpo_baseline_vs_novus_o...	nan	baseline_vs_observed	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	49	nan	nan	144.9687138737506	nan	nan	nan	2.0	ok	0	nan	0	0.6775706216442879	0.4816846028746816	0.5440430058842177	k_ar_diff=5	0	1	0	0	
comparison	comparison::silpo_baselin...	silpo_baseline_vs_novus_o...	silpo_baseline_vs_novus_o...	comparison_panel	silpo_baseline_vs_novus_o...	nan	baseline_vs_observed	linear	initial	daily	forward	ProducerUA	Retail	producer_prozorrr_retail_...	VECM	retail	producer+prozorro	49	nan	nan	0.463091036853965	nan	nan	nan	3.0	ok	0	nan	0	0.5099098773879907	0.9378049944694165	0.6508928482884466	k_ar_diff=1	0	1	0	0	
comparison	comparison::silpo_baselin...	silpo_baseline_vs_novus_o...	silpo_baseline_vs_novus_o...	comparison_panel	silpo_baseline_vs_novus_o...	nan	baseline_vs_observed	linear	initial	daily	forward	FarmGateUA	Retail	farmgate_producer_prozorrr...	VECM	retail	farm_gate+producer+prozorro	49	nan	nan	-19.03556384884796	nan	nan	nan	3.0	ok	0	nan	0	0.581945869295388	0.2197431710506238	0.7322400052328727	k_ar_diff=5	0	1	0	0	
comparison	comparison::silpo_novus_c...	silpo_novus_combined	silpo_novus_combined	comparison_panel	silpo_novus_combined	nan	observed	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	49	nan	nan	144.9687138737506	nan	nan	nan	2.0	ok	0	nan	0	0.6775706216442879	0.4816846028746816	0.5440430058842177	k_ar_diff=5	0	1	0	0	
comparison	comparison::silpo_novus_c...	silpo_novus_combined	silpo_novus_combined	comparison_panel	silpo_novus_combined	nan	observed	linear	initial	daily	forward	ProducerUA	Retail	producer_prozorrr_retail_...	VECM	retail	producer+prozorro	49	nan	nan	0.4755853663965547	nan	nan	nan	3.0	ok	0	nan	0	0.516367697550039	0.9320062462573419	0.6527988475863971	k_ar_diff=1	0	1	0	0	
comparison	comparison::silpo_novus_c...	silpo_novus_combined	silpo_novus_combined	comparison_panel	silpo_novus_combined	nan	observed	linear	initial	daily	forward	FarmGateUA	Retail	farmgate_producer_prozorrr...	VECM	retail	farm_gate+producer+prozorro	49	nan	nan	-20.8364357794366	nan	nan	nan	3.0	ok	0	nan	0	0.5751923476300824	0.243806286561618	0.7060092731291543	k_ar_diff=5	0	1	0	0	
brand	brand::Silpo:::cream::ob...	Silpo::cream	brand_panel	cream	Silpo	nan	observed	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	41	nan	nan	2.214812140355627	nan	nan	nan	2.0	ok	0	nan	0	0.9397102432548599	0.785228977301169	0.02977676690767751	k_ar_diff=5	0	1	0	0	
brand	brand::Silpo:::cream::ba...	Silpo::cream	brand_panel	cream	Silpo	nan	baseline	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	41	nan	nan	2.214812140355627	nan	nan	nan	2.0	ok	0	nan	0	0.9397102432548599	0.785228977301169	0.02977676690767751	k_ar_diff=5	0	1	0	0	
brand	brand::Silpo::alpro::hard...	Silpo:alpro:hard_cheese	brand_panel	hard_cheese	Silpo	alpro	observed	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	38	nan	nan	-3.281200511143566	nan	nan	nan	3.0	ok	0	nan	0	0.9908513112556221	0.5717024280852399	0.1445401016754244	k_ar_diff=5	0	1	0	0	
brand	brand::Silpo::alpro::hard...	Silpo:alpro:hard_cheese	brand_panel	hard_cheese	Silpo	alpro	observed	linear	initial	daily	forward	ProducerUA	Retail	producer_prozorrr_retail_...	VECM	retail	producer+prozorro	38	nan	nan	47.95088541122523	nan	nan	nan	2.0	ok	0	nan	0	0.9163638044284506	0.2547154811736901	0.6912454753178378	k_ar_diff=5	0	1	0	0	
brand	brand::Silpo::alpro::hard...	Silpo:alpro:hard_cheese	brand_panel	hard_cheese	Silpo	alpro	observed	linear	initial	daily	forward	FarmGateUA	Retail	farmgate_producer_prozorrr...	VECM	retail	farm_gate+producer+prozorro	38	nan	nan	-0.9714310925634219	nan	nan	nan	3.0	ok	0	nan	0	0.6725842844814571	0.3269936759872402	0.7790735901825621	k_ar_diff=5	0	1	0	0	
brand	brand::Silpo::alpro::hard...	Silpo:alpro:hard_cheese	brand_panel	hard_cheese	Silpo	alpro	baseline	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	38	nan	nan	-3.281200511143566	nan	nan	nan	3.0	ok	0	nan	0	0.9908513112556221	0.5717024280852399	0.1445401016754244	k_ar_diff=5	0	1	0	0	
brand	brand::Silpo::alpro::hard...	Silpo:alpro:hard_cheese	brand_panel	hard_cheese	Silpo	alpro	baseline	linear	initial	daily	forward	ProducerUA	Retail	producer_prozorrr_retail_...	VECM	retail	producer+prozorro	38	nan	nan	-7.760610186667271	nan	nan	nan	1.0	ok	0	nan	0	0.8735782124335273	0.7045780825430135	2.846144884425003e-25	k_ar_diff=5	0	1	0	0	
brand	brand::Silpo::alpro::hard...	Silpo:alpro:hard_cheese	brand_panel	hard_cheese	Silpo	alpro	baseline	linear	initial	daily	forward	FarmGateUA	Retail	farmgate_producer_prozorrr...	VECM	retail	farm_gate+producer+prozorro	38	nan	nan	-3.843326267827252	nan	nan	nan	2.0	ok	0	nan	0	0.4667970162627043	0.6101615234192457	0.6768170445553408	k_ar_diff=5	0	1	0	0	
brand	brand::Silpo:::milk::obs...	Silpo::milk	brand_panel	milk	Silpo	nan	observed	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	44	nan	nan	-5.244430544829129	nan	nan	nan	1.0	ok	0	nan	0	0.9312181151929981	0.9997841864116533	2.452346359994381e-156	k_ar_diff=5	0	1	0	0	
brand	brand::Silpo:::milk::bas...	Silpo::milk	brand_panel	milk	Silpo	nan	baseline	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	44	nan	nan	-5.244430544829129	nan	nan	nan	1.0	ok	0	nan	0	0.9312181151929981	0.9997841864116533	2.452346359994381e-156	k_ar_diff=5	0	1	0	0	
brand	brand::Silpo::alpro::sour...	Silpo:alpro:sour_cream	brand_panel	sour_cream	Silpo	alpro	observed	linear	initial	daily	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	38	nan	nan	-24.70966978176242	nan	nan	nan	2.0	ok	0	nan	0	0.1646681594712951	0.9828120691897677	0.9191199356810634	k_ar_diff=5	0	1	0	0	
brand	brand::Silpo::alpro::sour...	Silpo:alpro:sour_cream	brand_panel	sour_cream	Silpo	alpro	observed	linear	initial	daily	forward	ProducerUA	Retail	producer_prozorrr_retail_...	VECM	retail	producer+prozorro	38	nan	nan	-4.67154131655813	nan	nan	nan	1.0	ok	0	nan	0	0.948972335930743	0.9554569527195538	0.3166377522893951	k_ar_diff=5	0	1	0	0	
brand	brand::Silpo::alpro::sour...	Silpo:alpro:sour_cream																																				